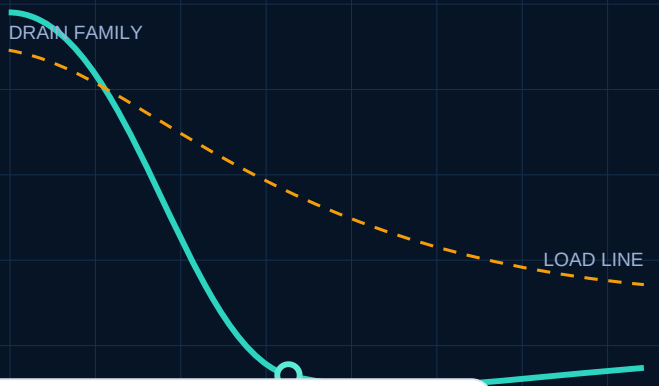
BASIC ELECTRONICS DEVICES · LABORATORY PROJECT

To Obtain the Drain and Transfer Characteristics of a MOSFET Connected in Common-Source Configuration

Interactive visualization and model-based analysis of an NMOS stage

BED LAB · COMMON-SOURCE CHARACTERIZATION · 2026



SUBMITTED BY	**Shivashish Das**
ENROLLMENT NO.	**12025052003049**
ROLL NO.	**A2-46**
DEPARTMENT	**Department of Electronics and Communication Engineering**
INSTITUTION	**Institute of Engineering and Management**
DATE OF SUBMISSION	**27 October 2026**

Submitted in partial fulfilment of the requirements of the Basic Electronics Devices laboratory.

ACKNOWLEDGMENT

I express my sincere gratitude to my project guide for the valuable guidance, patient review and constant encouragement extended throughout this laboratory project. I am also thankful to the faculty members of the **Department of Electronics and Communication Engineering** for providing the theoretical foundation and the facilities needed to complete the work. Finally, I thank my family and friends for their support and useful discussions during the preparation of this report.

ABSTRACT

The common-source (CS) configuration is the most widely used MOSFET stage for voltage amplification and is therefore a central experiment in an electronics laboratory. This project presents an interactive, browser-based visualization of the drain and transfer characteristics of an enhancement-mode NMOS transistor connected in the CS configuration. A Shichman–Hodges square-law model with channel-length modulation is used to compute the drain current over the cutoff, triode and saturation regions. The drain characteristics are obtained by sweeping V_{DS} for fixed values of V_{GS}, while the transfer characteristic is obtained by sweeping V_{GS} at a fixed V_{DS}. A DC load line is superimposed on the output family and its intersection with the device curve is solved numerically to locate the quiescent point. The visualization reports the operating region, transconductance, gate overdrive and power dissipation in real time, and allows the device parameters to be varied. For the nominal design (V_{DD} = 12 V, R_D = 2.2 kΩ, V_{th} = 2.0 V, k_n = 0.5 mA/V², λ = 0.02 V^{-1}), the quiescent point at V_{GS} = 4.0 V is (9.39 V, 1.19 mA) with g_m = 1.20 mS. The results reproduce the expected square-law behaviour and demonstrate how the load line and bias choice control the available output swing. The work is intended as a compact teaching aid that makes the link between device equations, measured-style curves and circuit design visible in one place.

Keywords: MOSFET, common-source amplifier, drain characteristics, transfer characteristics, quiescent point, transconductance, interactive visualization.

TABLE OF CONTENTS

1. Introduction . . . 4
1.1 Background and motivation . . . 4
1.2 Objectives . . . 4
1.3 Problem statement and scope . . . 4
1.4 Relevant theory and circuit . . . 4
2. Literature Review . . . 5
2.1 Device modelling . . . 5
2.2 Characterization practice . . . 6
2.3 Gap addressed by this work . . . 6
3. Methodology . . . 6
3.1 Device model and parameter selection . . . 6
3.2 Drain characteristics . . . 6
3.3 Transfer characteristics . . . 7
4. Implementation . . . 7
4.1 Circuit and numerical implementation . . . 7
4.2 Interactive visualization . . . 7
4.3 Report generation . . . 8
4.4 Challenges and solutions . . . 8
5. Results . . . 8
5.1 Drain characteristics . . . 8
5.2 Transfer characteristics . . . 9
5.3 Quiescent points and transconductance . . . 10
5.4 Summary of extracted metrics . . . 11
6. Discussion . . . 11
6.1 Interpretation . . . 11
6.2 Comparison with expected behaviour . . . 12
6.3 Limitations . . . 12
6.4 Potential improvements . . . 12
7. Conclusion . . . 12
8. References . . . 12

LIST OF FIGURES

Figure 1.1 NMOS common-source test circuit — p. 5

Figure 3.1 Characterization workflow — p. 7

Figure 4.1 Interactive visualization dashboard — p. 8

Figure 5.1 Drain characteristics with DC load line and quiescent point — p. 9

Figure 5.2 Transfer characteristics and transconductance tangent — p. 10

Figure 5.3 Transconductance versus gate–source voltage — p. 11

LIST OF TABLES

Table 3.1 Nominal model and bias parameters — p. 6

Table 5.1 Quiescent-point data for the output family — p. 10

Table 5.2 Summary of extracted performance metrics — p. 11

1. Introduction

1.1 Background and motivation

The metal–oxide–semiconductor field-effect transistor (MOSFET) is the dominant active device in modern analog and digital integrated circuits. Its insulated gate draws almost no steady-state current, so the device can be controlled by voltage rather than current. When the source terminal is shared by the input and output loops, the transistor is said to be in the common-source (CS) configuration. The gate–source voltage sets the channel conductivity, the drain–source voltage determines the operating region, and a drain resistor converts the drain current into an output voltage. This is the basic building block of voltage amplifiers, active loads, current sources, switches and many sensor front ends.

Characterizing the device is therefore an essential first step in analog design. The *drain* or *output* characteristic plots I_D against V_{DS} for several fixed values of V_{GS}, while the *transfer* characteristic plots I_D against V_{GS} at a fixed V_{DS}. Together they reveal the threshold voltage, the transconductance parameter, the onset of saturation, the output resistance and the effect of channel-length modulation. In a conventional laboratory these curves are traced point by point using two supplies and a pair of meters, or with a semiconductor parameter analyzer. The measurement is instructive but time-consuming, and the resulting plots are usually static.

This project addresses that limitation by combining the standard device equations with an interactive web visualization. The user can change the gate bias, supply voltage, load resistor, threshold voltage, transconductance parameter and channel-length modulation, and immediately see the curves, the load line and the quiescent point respond. The same model is then used to produce a compact laboratory report with reproducible figures and tabulated results.

1.2 Objectives

- Model an enhancement-mode NMOS transistor operating in the common-source configuration using the Shichman–Hodges square-law equations.
- Obtain the drain characteristics (I_D versus V_{DS}) for a family of gate–source voltages.
- Obtain the transfer characteristic (I_D versus V_{GS}) at a fixed drain–source voltage in saturation.
- Locate the quiescent point from the intersection of the device curve and the DC load line, and extract V_{th}, k_n and g_m.
- Build an interactive visualization and document the method, results and limitations in a laboratory report.

1.3 Problem statement and scope

The problem is to characterize a MOSFET in the CS configuration and to make the relationship between the device equations, the measured-style curves and the circuit bias easy to explore. The scope of this work is limited to a single enhancement-mode NMOS device under DC bias at room temperature. The model includes the square-law channel, channel-length modulation and a resistive drain load, but it does not include subthreshold conduction, mobility degradation, temperature dependence, parasitic capacitances or high-frequency effects. The numerical values are representative of a small-signal NMOS and are intended for teaching and visualization rather than for predicting a specific part.

1.4 Relevant theory and circuit

Figure 1.1 shows the test circuit. The source is grounded, the gate is biased at V_{GS} and the drain is connected to the supply through R_D. The drain current is measured in the drain branch and the drain–source voltage is measured across the device. Depending on the terminal voltages, the MOSFET operates in cutoff, triode or saturation.

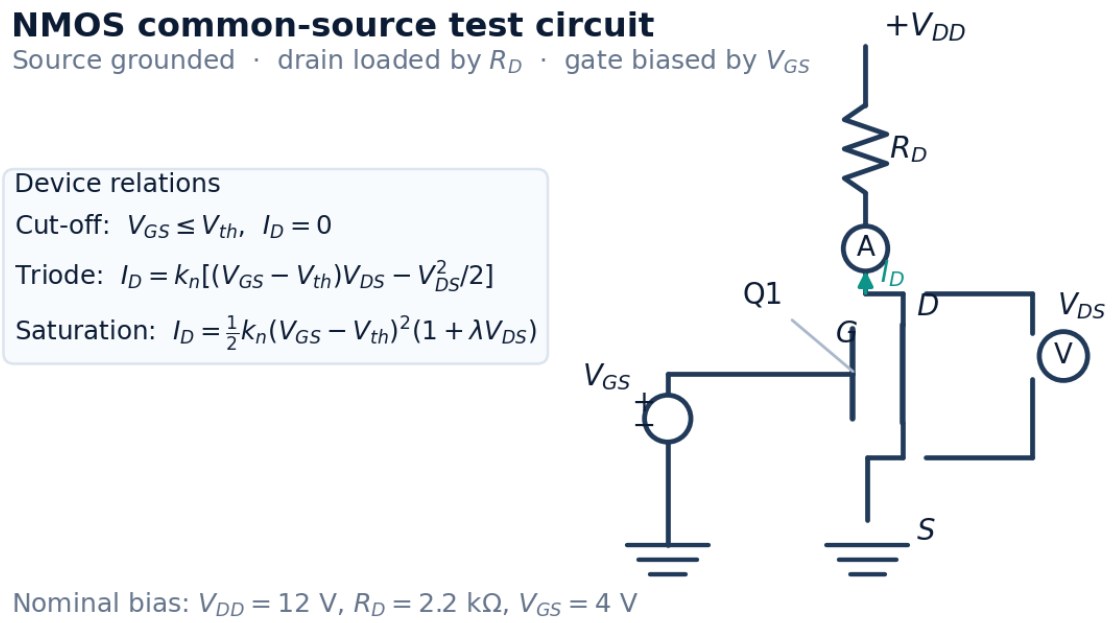


Figure 1.1. NMOS common-source test circuit and the three regions of operation.

In cutoff, $V_{GS} \le V_{th}$ and no inversion channel is formed:

$$I_D = 0 \qquad (V_{GS} \le V_{th}) \qquad (1)$$

In triode, the channel is continuous and the device behaves as a voltage-controlled resistor:

$$I_D = k_n[(V_{GS} - V_{th})V_{DS} - \tfrac{1}{2}V_{DS}^2](1 + \lambda V_{DS}) \qquad (V_{DS} < V_{GS} - V_{th}) \qquad (2)$$

In saturation, the channel pinches off near the drain and the current is approximately independent of V_{DS}:

$$I_D = \tfrac{1}{2}k_n(V_{GS} - V_{th})^2(1 + \lambda V_{DS}) \qquad (V_{DS} \ge V_{GS} - V_{th}) \qquad (3)$$

The small-signal transconductance in saturation is the slope of the transfer characteristic: $g_m = \partial I_D/\partial V_{GS} = k_n(V_{GS} - V_{th})(1 + \lambda V_{DS})$. The external circuit imposes the DC load line $I_D = (V_{DD} - V_{DS})/R_D$, and its intersection with the device curve is the quiescent point Q.

2. Literature Review

2.1 Device modelling

The square-law MOSFET model originates from the gradual-channel approximation and was popularized for circuit simulation by Shichman and Hodges in 1968. It expresses the drain current in terms of the gate overdrive and the drain–source voltage using a small number of parameters: threshold voltage, transconductance parameter and channel-length modulation. The model is the basis of the Level-1 MOSFET model in SPICE and remains the first model taught in most electronics courses because it gives closed-form expressions for the operating point, transconductance and output resistance. More advanced models such as BSIM add short-channel effects, subthreshold conduction, mobility degradation and charge conservation, but they obscure the basic device behaviour that a first laboratory is intended to demonstrate.

2.2 Characterization practice

Traditional characterization uses a curve tracer or a source-measure unit to sweep one terminal while holding the others fixed. The resulting family of curves is read manually to extract V_{th}, k_n and the saturation slope. Modern automated setups use LabVIEW or Python to control the instrument and plot the data. Circuit simulators such as LTspice, ngspice and Multisim are widely used for pre-lab work, and educational web applets such as Falstad's circuit simulator provide immediate visual feedback. These tools are powerful, but the full simulators have a steep learning curve for a first-year student, while many applets show only a single curve or hide the underlying equations.

2.3 Gap addressed by this work

There is a useful middle ground between a full SPICE deck and a static textbook plot: a small, transparent model that exposes the equations and lets the student manipulate the bias directly. The present work fills that gap with a dependency-free web page that plots the drain and transfer characteristics, superimposes the load line, solves for the quiescent point and reports the small-signal parameters. The same equations are used to generate the figures in this report, so the visualization and the written analysis remain consistent.

3. Methodology

3.1 Device model and parameter selection

The NMOS device is represented by the Shichman–Hodges equations (1)–(3). The nominal parameters were chosen to resemble a small enhancement-mode NMOS suitable for a 12 V laboratory supply. The threshold voltage is 2.0 V, the transconductance parameter is 0.5 mA/V² and the channel-length modulation coefficient is 0.02 V^{-1}. The drain resistor is 2.2 kΩ and the supply is 12 V. These values place the quiescent point comfortably inside saturation at V_{GS} = 4 V and allow the entire output family to be displayed on a single graph. Table 3.1 lists the parameters.

Parameter	Symbol	Value	Unit
Threshold voltage	V_{th}	2.0	V
Transconductance parameter	k_n	0.50	mA/V²
Channel-length modulation	λ	0.020	V^{-1}
Supply voltage	V_{DD}	12	V
Drain resistor	R_D	2.2	kΩ
Gate–source bias at Q	V_{GS}	4.0	V
Drain–source voltage for transfer	V_{DS}	10	V

Table 3.1. Nominal model and bias parameters used throughout the work.

3.2 Drain characteristics

For the output family, V_{GS} is held at each of eight values from 2.5 V to 6.0 V in 0.5 V steps. For every value, V_{DS} is swept from 0 V to V_{DD} in small increments and I_D is evaluated from equations (1)–(3). The resulting curves are plotted on a common set of axes. A DC load line is drawn from (0, V_{DD}/R_D) to (V_{DD}, 0). The quiescent point for the selected gate bias is found by solving $I_D(V_{GS}, V_{DS}) = (V_{DD} - V_{DS})/R_D$ with a bisection search. Because the device current increases with V_{DS} in saturation while the load-line current decreases, the root is unique.

3.3 Transfer characteristics

For the transfer characteristic, V_{DS} is fixed at 10 V, which keeps the device in saturation for all gate voltages above threshold. V_{GS} is swept from 0 V to 6 V and I_D is evaluated from the saturation equation. The threshold voltage is read from the turn-on point, and k_n is obtained from the square-law fit. The transconductance is computed by a central difference of the transfer curve, which avoids relying on the analytic derivative and therefore provides an independent numerical check. Figure 3.1 summarizes the workflow.

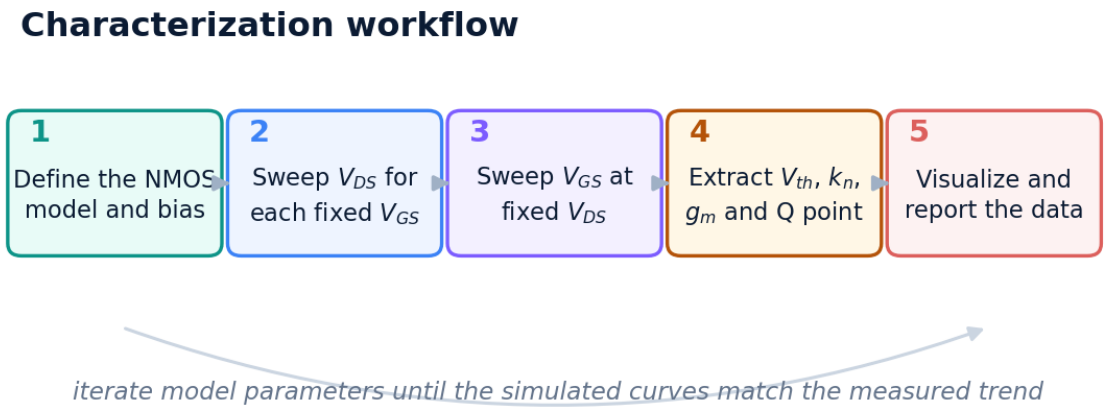


Figure 3.1. Characterization workflow from model definition to visualization and reporting.

4. Implementation

4.1 Circuit and numerical implementation

The test circuit follows the standard CS topology: the source is grounded, the gate is driven by a variable DC source and the drain is connected to V_{DD} through R_D. The model is implemented as a small JavaScript function that returns the drain current for any pair of terminal voltages. The output characteristics are produced by evaluating this function on a dense grid of V_{DS} values, and the transfer characteristic is produced by evaluating it on a dense grid of V_{GS} values. The quiescent point is found with 70 bisection iterations, which is more than sufficient for the precision displayed in the interface.

```
function drainCurrent(vgs, vds) { const vov = vgs - vth; if (vov <= 0) return 0; // cutoff const f =
1 + lambda * vds; if (vds < vov) // triode return kn * (vov * vds - 0.5 * vds * vds) * f; return 0.5
* kn * vov * vov * f; // saturation }
```

Listing 1. Core of the square-law NMOS model used by the visualization.

4.2 Interactive visualization

The front end is a single-page application built with HTML, CSS and vanilla JavaScript. The characteristic curves are drawn as SVG paths, so they remain sharp at any zoom level and can be styled with the rest of the design. The control panel exposes V_{GS}, V_{DD}, R_D, V_{th}, k_n and λ. Changing any slider re-computes the curves, the load line, the quiescent point, the operating region and the tabulated data immediately. A tab switches between the drain and transfer views; a sweep button animates V_{GS}; a tooltip reads out the current at the cursor; and the table can be copied or exported as CSV. Figure 4.1 shows the main workspace.

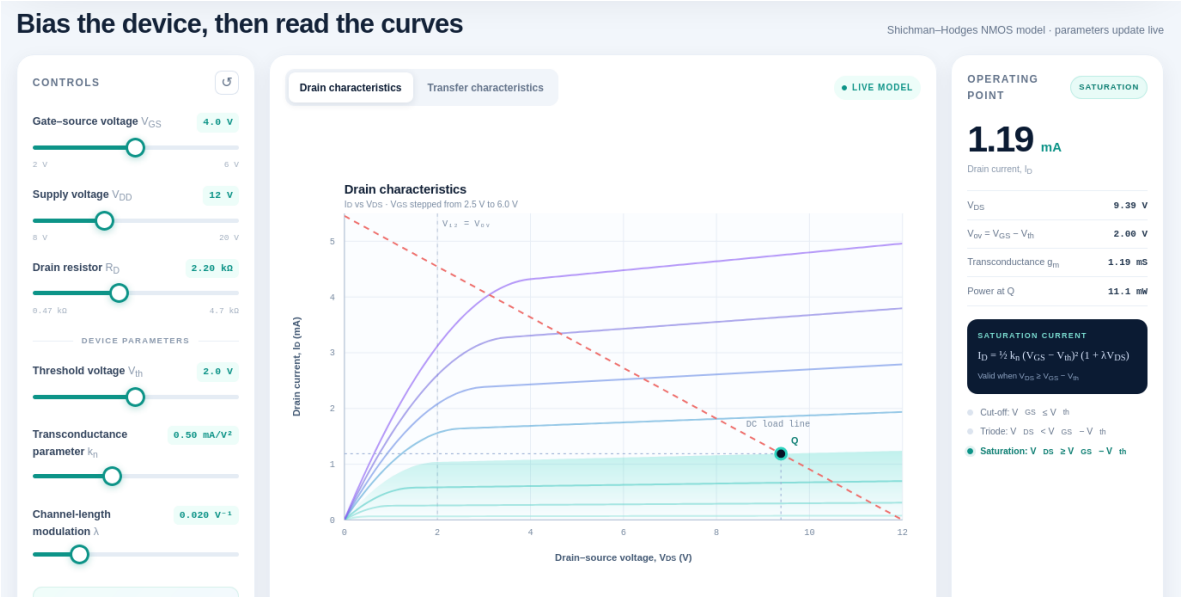


Figure 4.1. Interactive visualization dashboard: bias controls, characteristic plot and live operating-point readout.

4.3 Report generation

The figures in this report are produced by a Python script using NumPy and Matplotlib, and the document itself is assembled with ReportLab. A single parameter set is shared by the web application and the figure generator, so the numerical values in the text, tables and plots agree. The report is built from the same model rather than from manually transcribed numbers, which reduces the risk of inconsistency between the visualization and the written analysis.

4.4 Challenges and solutions

Three practical difficulties arose during implementation. First, the SVG chart is re-rendered on every slider movement; drawing the full family at a high sampling density made the interaction sluggish, so the number of samples was reduced and the curves are clipped to the plot area. Second, the load-line intersection must be found even when the operating point moves from saturation into triode; a simple analytic solution is not valid across all regions, so a bracketed bisection search was used instead. Third, the chart labels must remain readable when the device parameters change the current scale; the axis maximum is chosen from the data with a small headroom and the ticks are generated from a "nice number" rule. These choices keep the interface responsive and the plots well scaled.

5. Results

5.1 Drain characteristics

Figure 5.1 shows the output family for V_{GS} = 2.5 V to 6.0 V. Each curve rises almost linearly for small V_{DS}, bends over at $V_{DS} \approx V_{GS} - V_{th}$, and then becomes nearly horizontal. The nearly flat region is saturation; its small positive slope is caused by channel-length modulation. The curve for V_{GS} = 2.5 V has the smallest overdrive and therefore the lowest saturation current, while the curve for V_{GS} = 6.0 V has the largest. The dashed red line is the DC load line for R_D = 2.2 kΩ; its intersection with the V_{GS} = 4.0 V curve is the quiescent point Q = (9.39 V, 1.19 mA). At this bias the transistor is in saturation, as required for a linear amplifier.

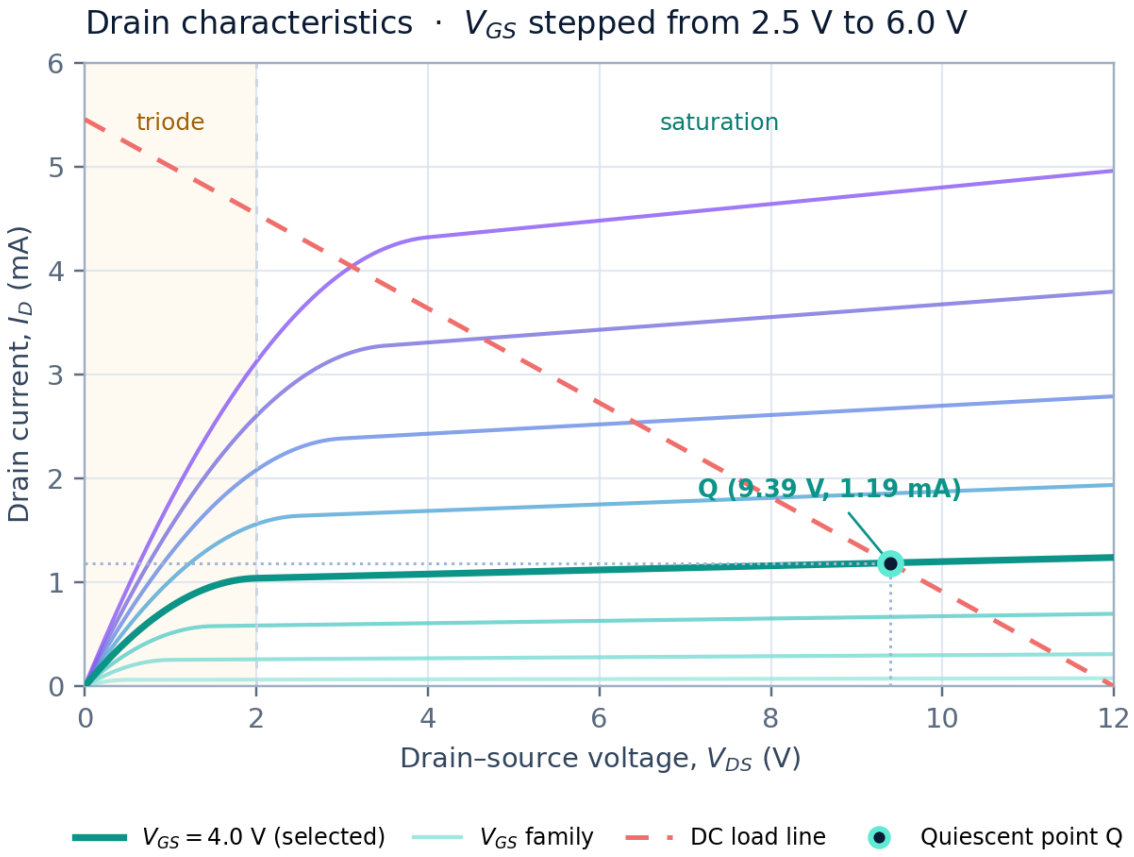


Figure 5.1. Drain characteristics for VGS = 2.5 V to 6.0 V, with the DC load line and the quiescent point at VGS = 4.0 V.

5.2 Transfer characteristics

The transfer characteristic in Figure 5.2 was obtained at V_{DS} = 10 V. The current is essentially zero below V_{th} = 2.0 V and follows a square law above threshold. The extracted threshold voltage is 2.0 V and the curve passes through 1.20 mA at V_{GS} = 4.0 V. The dashed amber line is the tangent at the quiescent point; its slope is the transconductance, g_m = 1.20 mS. The tangent is a good local approximation for small signals around the bias point, but it deviates from the curve for large excursions because the transfer characteristic is quadratic rather than linear.

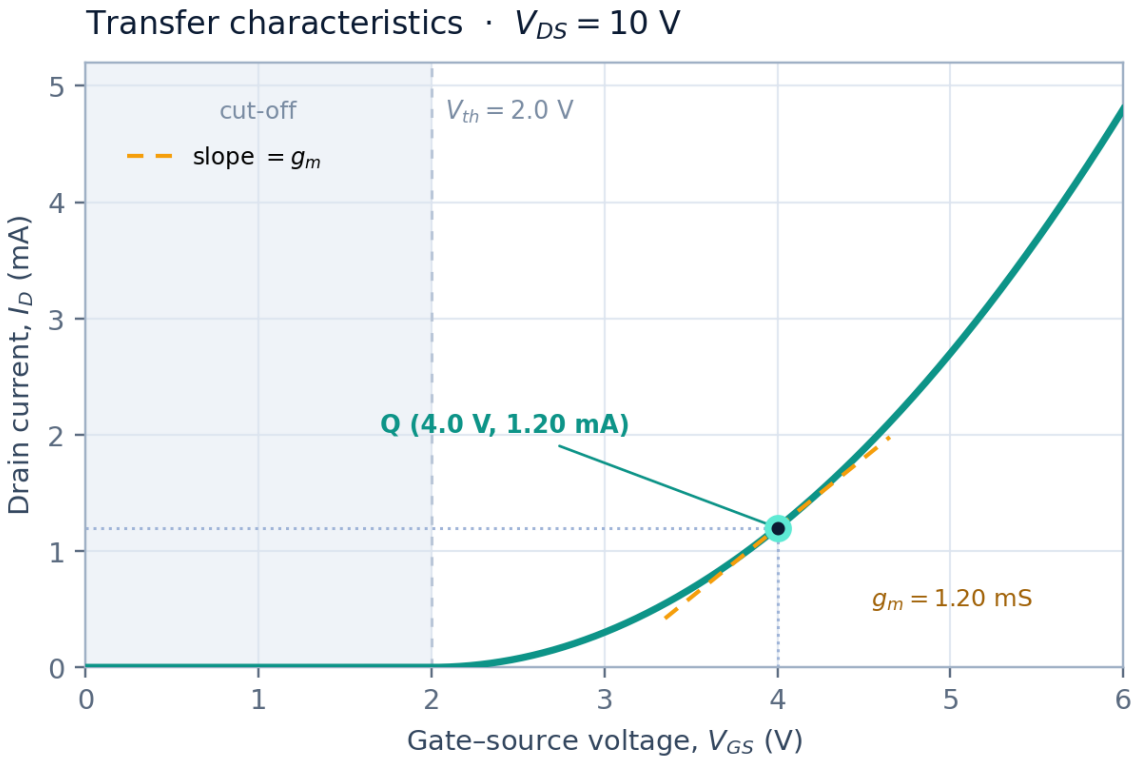


Figure 5.2. Transfer characteristic at VDS = 10 V, showing the threshold voltage and the tangent used to extract gm.

5.3 Quiescent points and transconductance

Table 5.1 lists the quiescent point for each gate voltage in the family. As V_{GS} increases, the drain current rises and the drain voltage falls along the load line. For V_{GS} = 6.0 V, the operating point moves into the triode region because the required current is larger than the saturation current available at the resulting drain voltage. This is a useful design observation: a larger gate drive does not always keep the device in saturation when a resistive load is present.

V_{GS} (V)	$V_{DS,Q}$ (V)	$I_{D,Q}$ (mA)	Region	g_m (mS)
2.5	11.83	0.08	Saturation	0.31
3.0	11.33	0.31	Saturation	0.61
3.5	10.50	0.68	Saturation	0.91
4.0	9.39	1.19	Saturation	1.19
4.5	8.01	1.81	Saturation	1.45
5.0	6.41	2.54	Saturation	1.69
5.5	4.64	3.35	Saturation	1.91
6.0	3.11	4.04	Triode	1.65

Table 5.1. Quiescent-point data for the output family (VDD = 12 V, RD = 2.2 kΩ).

Figure 5.3 plots the transconductance in saturation against gate–source voltage. In the square-law model g_m is proportional to the gate overdrive, so the plot is a straight line that starts at V_{th}. The value at the quiescent point is 1.20 mS. The numerical derivative used for this plot agrees with the analytic expression to the precision shown, which confirms that the implementation and the small-signal interpretation are consistent.

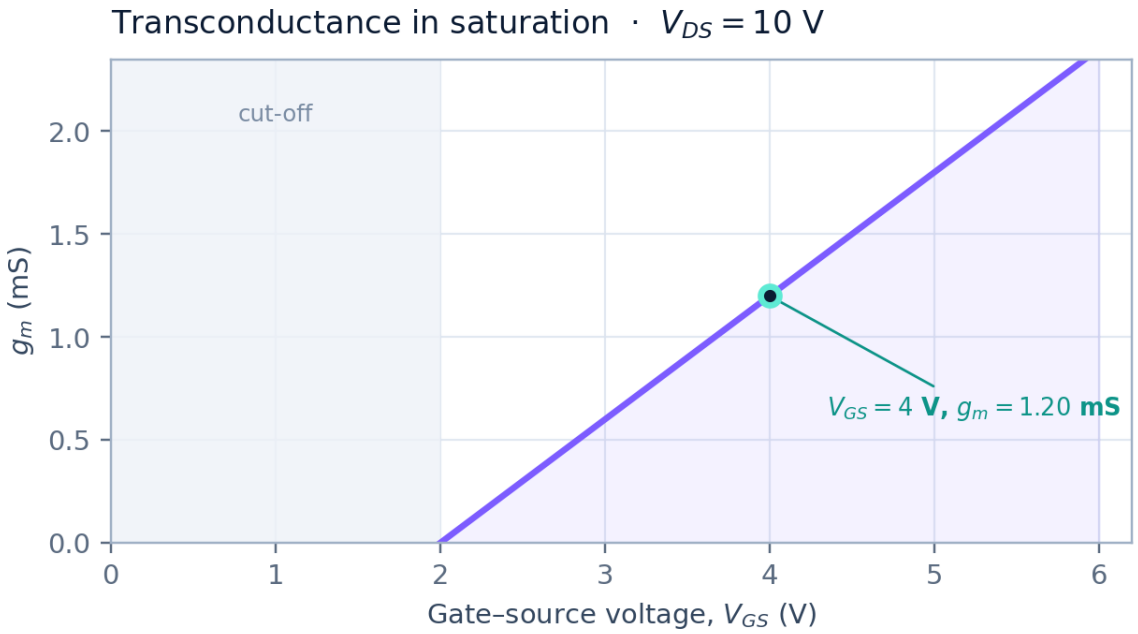


Figure 5.3. Transconductance versus gate–source voltage in saturation; the quiescent point is marked.

5.4 Summary of extracted metrics

Metric	Value	How obtained
Threshold voltage, V_{th}	2.0 V	Turn-on point of the transfer curve
Transconductance parameter, k_n	0.50 mA/V²	Square-law fit above threshold
Quiescent current, $I_{D,Q}$	1.19 mA	Load-line intersection at V_{GS} = 4 V
Quiescent voltage, $V_{DS,Q}$	9.39 V	Load-line intersection at V_{GS} = 4 V
Transconductance at Q, g_m	1.20 mS	Slope of transfer curve at Q
Output resistance, $r_o = 1/(\lambda I_D)$	≈ 42 kΩ	Channel-length modulation parameter

Table 5.2. Summary of the extracted device and bias metrics at the nominal operating point.

The results are internally consistent: the same threshold voltage, transconductance parameter and channel-length modulation coefficient reproduce the output family, the transfer curve and the quiescent points. The small difference between the analytic g_m and the numerical slope is below the display precision. The output resistance is approximate because λ is treated as a constant in the model, whereas in a real device it varies with bias.

6. Discussion

6.1 Interpretation

The curves show the three expected regions of MOSFET operation. In the triode region the channel behaves as a voltage-controlled resistor, so the current rises with V_{DS}. As V_{DS} approaches the overdrive voltage, the channel pinches off near the drain and the current saturates. The residual slope in saturation is the channel-length modulation, which gives the device a finite output resistance and therefore limits the maximum voltage gain of a simple CS stage. The load line makes the bias choice explicit: the quiescent point is fixed by both the device and the external circuit, not by the gate voltage alone.

6.2 Comparison with expected behaviour

The square-law model predicts that the saturation current should scale with the square of the gate overdrive. The values in Table 5.1 follow this trend closely: doubling the overdrive from 0.5 V to 1.0 V increases the current by roughly a factor of four, and the same scaling is visible in the transfer curve. The transconductance is linear in the overdrive, as expected from $g_m = k_n(V_{GS} - V_{th})(1 + \lambda V_{DS})$. These trends match the textbook treatment of the Level-1 MOSFET model and the behaviour of a real device at moderate current levels, although real short-channel devices show a softer turn-on and a more pronounced output conductance.

6.3 Limitations

- The model is generic and is not fitted to a measured device; the numerical values are representative rather than commercial.
- Subthreshold conduction, mobility degradation, temperature effects and parasitic capacitances are not included.
- The DC analysis does not predict bandwidth, slew rate or high-frequency gain.
- The load is a single resistor; active loads and current-source biasing are outside the scope of this experiment.

6.4 Potential improvements

Future versions could import measured I–V data and fit V_{th}, k_n and λ directly from the measurements, add a temperature-dependent mobility model, and include a small-signal AC analysis so that the gain and bandwidth of the CS stage can be explored. A graphical load-line tool that lets the user drag R_D and V_{DD} while watching the available output swing would make the design trade-off even more intuitive. Finally, the visualization could be extended to a PMOS device and to a CMOS inverter, which would connect the experiment to digital circuit design.

7. Conclusion

This project set out to obtain and visualize the drain and transfer characteristics of an NMOS transistor connected in the common-source configuration. A Shichman–Hodges model was implemented in an interactive web application and used to generate the output family, the transfer curve, the DC load line and the quiescent point. For the nominal design, the device turns on at 2.0 V, delivers 1.19 mA at $V_{DS} = 9.39$ V when $V_{GS} = 4.0$ V, and has a transconductance of 1.20 mS at that bias. The extracted trends follow the expected square-law behaviour, and the load line explains why the operating point moves from saturation to triode as the gate drive is increased. The objectives of the experiment were therefore met. The main open questions concern real-device effects: fitting the model to measured data and extending the analysis to temperature, short-channel effects and high-frequency behaviour would make the tool more useful for practical design.

8. References

[1] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed. Hoboken, NJ, USA: Wiley, 2007.

[2] H. Shichman and D. A. Hodges, “Modeling and simulation of insulated-gate field-effect transistor switching circuits,” *IEEE Journal of Solid-State Circuits*, vol. SC-3, no. 3, pp. 285–289, Sep. 1968.

[3] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 8th ed. New York, NY, USA: Oxford University Press, 2019.

[4] B. Razavi, *Fundamentals of Microelectronics*, 3rd ed. Hoboken, NJ, USA: Wiley, 2021.

[5] R. L. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed. Boston, MA, USA: Pearson, 2013.

[6] M. H. Rashid, *Microelectronic Circuits: Analysis and Design*, 3rd ed. Boston, MA, USA: Cengage Learning, 2016.

[7] Analog Devices, “MOSFET characterization using the 4200A-SCS parameter analyzer,” Application Note, 2020.

[8] Analog Devices, “LTspice simulator,” 2024. [Online]. Available: https://www.analog.com/ltspice